# Past the Knee: Rate–Distortion-Aware Comparison of Tensor-Structured Dictionaries, and an Atoms-per-Parameter Law

Technical Report — Extension to [1]

July 30, 2026

**Abstract**

A companion report [1] placed five tensor-structured dictionaries (CP, Tucker, Tensor-Train, Tensor-Ring, Block-Term) and an unstructured baseline in a single harness and compared them at a matched parameter budget on a hyperspectral cube, concluding that the unstructured baseline wins at $\approx$5000 parameters at both 6 and 18 dB and that the ranking does not flip with noise. This extension re-examines that conclusion and finds it is an artifact of *where on the rate–distortion curve the comparison was read*. We show that at every point of the original budget grid, and for every method, the rate–distortion slope $\partial\,\mathrm{PSNR}/\partial\,\mathrm{params}$ is strictly negative: the entire sweep lies past the knee, so the matched slice measures which family degrades most gracefully under forced overspending rather than which prior fits the data. Extending the grid downward by a factor of seven inverts the ranking. Tensor-Train leads at 6 dB with 4.8$\times$ fewer parameters than the baseline's optimum (36.64 vs. 35.45 dB), and at 18 dB the baseline is *last* of six (paired inference later shows the structured leaders statistically tied — the baseline's position is what is significant). We then show that the sign of that slope is a *regime diagnostic*. On natural RGB patches every slope is positive and a single scalar — atoms purchased per parameter, $\alpha = K/P_{\mathrm{dict}}$ — rank-orders the six families perfectly at both noise levels (Spearman $\rho = +1.00$), and also predicts the ranking published in [1] for corrected Tucker at $\rho = +0.94$. On the hyperspectral cube, where slopes are negative, the same scalar anti-correlates ($\rho = -0.83$ at 6 dB). We give a two-regime account that reconciles these: the parameter budget conflates atom count with per-atom expressiveness, and the slope sign determines which of the two is the binding resource. Three secondary corrections are reported: a reachability analysis showing a single Kronecker dictionary cannot spend more than 328 of the nominal 5000 parameters under a 600-atom cap; a jointly fitted multi-block Kronecker dictionary that improves on residual boosting by 2–5 dB without changing the conclusion; and a variance decomposition showing dataset draw dominates initialization roughly 5:1, so that paired testing is the appropriate instrument. A second round of experiments then tests the framework itself. A *registered* prediction from $\alpha$ alone, for a family never previously run (a low-rank-unfolding format), lands exactly at 6 dB (30.90 predicted, 30.90 observed, correct rank) and within 0.22 dB at 18 dB. An inpainting study locates the law's boundary: with 50% of entries hidden, $\alpha$'s perfect ordering collapses ($\rho = +1.00 \to +0.33$) while the structured prior's advantage at the knee grows — $\alpha$ governs denoising capacity, not extrapolation. Paired inference at the optima softens our own point-estimate ranking to its defensible form: the structured families are statistically tied at the knee, and only the baseline's last place is significant ($p \leq 0.007$). Releasing the atom cap yields a new overall RGB record (36.22 dB from 548 parameters, a 4-block Kronecker union), and on *real* hyperspectral data (Indian Pines) the synthetic cube's regime does not transfer: the task is atom-hungry, the separable dictionary wins everything, and the baseline is last everywhere.

# 1 Introduction

The companion report [1] makes a methodological argument that we endorse: because an observed gap between two dictionary families may reflect the optimizer or hand-picked hyperparameters rather than the decomposition, families should be compared inside a shared harness at a matched cost. It then instantiates that argument with a matched-parameter-budget rate–distortion study on a hyperspectral (HSI) cube, sweeping a budget $B \in \{2000, 5000, 9000\}$ and reading a ranking off the vertical slice at $B \approx 5000$.

This extension accepts the harness and the fairness argument, and interrogates the instrument. A matched-budget slice is a one-dimensional cut through a two-dimensional surface, and its verdict depends on *where* the cut is taken. Our central observation is elementary and, we believe, consequential: in [1], every rate–distortion curve at both noise levels is monotonically *decreasing* across the entire sampled budget range. A curve that only ever falls has its optimum outside the sampled window, to the left. The reported ranking is therefore a ranking of tail behaviour.

**Contributions.**

1. **A slope diagnostic and a corrected sweep** (Sections 3, 4). We show all 36 curve segments in the original HSI study have negative slope, extend the budget grid down to 300 parameters, remove two clamps in the budget-to-rank map that prevented families from reaching their nominal budgets, and re-run. The ranking inverts at both noise levels (Section 5.1).

2. **An atoms-per-parameter law, and its domain of validity** (Section 5.3). On natural RGB patches, $\alpha = K/P_{\text{dict}}$ evaluated at each family's own optimum rank-orders test PSNR perfectly at both noise levels ($\rho = +1.00$), and reproduces the ranking that [1] publishes for corrected Tucker ($\rho = +0.94$). On the HSI cube the same scalar anti-correlates. We identify the slope sign as the variable that separates the two regimes.

3. **Registered predictions and external validation** (Section 6). A second experimental round tests the framework prospectively: a registered $\alpha$ prediction for a never-run family, an inpainting study that locates the law's domain boundary, paired inference at the optima, a cap-release study, a structure-mix phase sweep, a convergence ablation, and evaluations on a three-image RGB corpus and on real hyperspectral data (Indian Pines).

4. **Three corrections to the budget protocol** (Sections 3.2, 5.5, 5.6): a reachability analysis of Kronecker dictionaries under an atom cap; a jointly fitted multi-block Kronecker dictionary replacing residual boosting; and a variance decomposition motivating paired inference.

We stress at the outset that this is a *sharpening* of [1], not a refutation of its programme. Two of its own conclusions are strengthened here: the regime-dependence thesis becomes quantitative, and the observation that "matched-parameter is not matched-atom" becomes the organizing principle rather than a caveat.

# 2 Setup

## 2.1 Harness and reproduction fidelity

We adopt the harness of [1] unchanged: OMP sparse coding against a materialized effective dictionary $D$, MOD/K-SVD-style atom updates, per-atom projection onto the relevant low-rank manifold

for CP/TT/TR/BTD, and separable alternating least squares for the Kronecker ("Tucker") dictionary. Parameter accounting follows [1, Table 1] exactly.

The dictionary-learning module of [1] was not available to us, so it was re-implemented from its published interface and parameter-cost formulas. Table 1 reports agreement on the original HSI budget sweep. Five of six families reproduce to within 0.19 dB at every budget and both noise levels, with mean absolute deviation 0.09 dB. Tucker is the exception and is discussed in Section 3.2; our separable ALS attains 32.90 dB where [1] reports 26.90 dB at comparable size, a discrepancy consistent with the optimizer sensitivity that [1] itself documents in its Section 6.3 fairness correction.

| | 6 dB | | | 18 dB | | |
|---|---|---|---|---|---|---|
| Family | $B$=2000 | $B$=5000 | $B$=9000 | $B$=2000 | $B$=5000 | $B$=9000 |
| Vectorized | +0.08 | −0.03 | −0.08 | +0.05 | +0.22 | +0.13 |
| CP | +0.01 | −0.02 | +0.16 | +0.00 | +0.02 | −0.44 |
| TT | −0.01 | −0.16 | −0.19 | −0.13 | −0.12 | −0.24 |
| TR | −0.19 | −0.18 | −0.17 | −0.08 | +0.26 | −0.16 |
| BTD | +0.07 | −0.05 | +0.05 | −0.09 | −0.14 | −0.22 |
| Tucker | +6.00 | +6.00 | +6.00 | +5.17 | +5.17 | +5.17 |

**Table 1:** Reproduction fidelity: our PSNR minus that of [1, Fig. 9 data], in dB. Mean absolute deviation excluding Tucker is 0.09 dB. Tucker is discussed in Section 3.2.

### 2.2 Data

Two datasets are used. **HSI** is the synthetic cube of [1]: patches of shape $6 \times 6 \times 16$ drawn from a mixture of five smooth Gaussian endmember spectra modulated by smooth spatial abundance maps, mean-removed and unit-normalized, corrupted to a target SNR. **RGB** is $8 \times 8 \times 3$ patches from the standard `astronaut` photograph, with the same normalization and noise model. We add a validation split to the RGB generator, which the original lacked. Throughout, 300 training, 100 validation and 100 test patches; training sparsity 4; encoding sparsity 8; two seeds unless stated.

## 3 Why a matched-budget slice can invert a ranking

### 3.1 The slope diagnostic

Let $\mathrm{P}_f(B)$ denote the test PSNR of family $f$ at parameter budget $B$. A rate–distortion curve for a denoising task is generically unimodal: too few parameters underfits, too many fits the noise. A matched-budget comparison is informative about modelling quality only if the sampled window brackets the optimum, i.e. if the sign of $\partial \mathrm{P}_f / \partial B$ varies across the window.

In [1, Fig. 9] it does not. Across 6 families × 2 noise levels × 2 budget intervals, every finite difference is negative. Aggregated from smallest to largest budget, the total change is −5.1 dB (Vectorized), −6.9 (CP), −8.7 (TT), −7.9 (TR) and −7.4 (BTD) at 6 dB, and −0.3, −6.1, −5.1, −3.6, −5.8 at 18 dB. Every family is on its descending branch throughout.

The consequence is that the 5000-parameter slice ranks families by *how slowly they decay when forced to overspend*. That quantity has a mechanical explanation which is not about structure at all: a family that buys few expensive atoms adds capacity in coarse increments, so at a fixed excess budget it holds fewer effective degrees of freedom than a family that buys many cheap atoms. The unstructured baseline, whose atoms cost $\prod_i n_i = 576$ parameters each, is the extreme case — at

$B = 5000$ it holds $K = 9$ atoms against Tensor-Train's 42. It is therefore *expected* to lead on a descending branch, independently of whether its prior suits the data.

### 3.2 Reachability: two clamps in the budget-to-rank map

The budget is converted to an atom count by $K = \text{clip}(\text{round}(B/c_f), K_{\min}, K_{\max})$ with $K_{\min} = 8$, $K_{\max} = 600$, where $c_f$ is the per-atom cost of family $f$. Both clamps bind, and they bind asymmetrically.

**The floor.** A vectorized atom costs 576 parameters, so the cheapest admissible unstructured dictionary holds $8 \times 576 = 4608$ parameters. At the nominal $B = 2000$ budget the baseline in [1] in fact spends 4608 — a 2.3× overspend. The $B = 2000$ row is therefore not a 2000-parameter comparison.

**The cap.** For the Kronecker dictionary the cost is $\sum_i n_i k_i$ while the atom count is $\prod_i k_i$. Cost grows linearly in the mode ranks and atom count grows multiplicatively, so a cap on atoms binds long before cost approaches the budget. Enumerating all $(k_1, k_2, k_3)$ with $\prod_i k_i \leq 600$ for $n = (6, 6, 16)$ gives a maximum of 328 parameters, at $(6, 6, 16)$ with 576 atoms (Figure 4, left). Tucker is thus 93.4% under budget at the 5000-parameter slice and returns an identical model at all three budgets, which is visible in the original data as three repeated entries. Reaching 5000 parameters with a single Kronecker dictionary would require $k_i \approx 10^2$ and hence $\sim 10^6$ atoms.

### 3.3 Dictionary parameters are not the total description length

The budget charges for $D$ but not for the codes. Storing a code costs $s$ values plus $s \log_2 K$ bits of support, so a per-patch description length in float-equivalents is

$$\text{DL}_f \;=\; \frac{P_{\text{dict}}}{M} \;+\; \frac{s\,(b + \log_2 K)}{b}, \qquad b = 32. \tag{1}$$

At $B = 5000$ the baseline holds $K = 9$ and TT holds $K = 42$, a difference of $s \log_2(42/9) \approx 9$ bits per patch. Charging only for the dictionary therefore systematically favours few-expensive-atom designs — the same families favoured by the descending branch. The two biases compound rather than cancel.

## 4 Corrected protocol

We make three changes and hold everything else fixed.

1. **Knee-seeking budget grid.** Extend the grid to $B \in \{300, 600, 1200, 2000, 5000, 9000\}$ on HSI and $B \in \{400, \ldots, 25600\}$ on RGB, and lower the atom floor to $K_{\min} = 2$. Report each family at its *own* argmax over $B$ in addition to the fixed slice.

2. **Reachability-aware Kronecker budget.** Because extra Kronecker *blocks* scale cost and atom count both linearly, a $b$-block union is the only construction by which a separable dictionary can legally spend a large budget. We fit it by block-coordinate descent on the union dictionary rather than by residual boosting (Section 5.5).

3. **Paired inference.** Separate the dataset seed from the initialization seed, replicate over $5 \times 2$, and compare families by paired tests on shared dataset draws (Section 5.6).

# 5 Results

## 5.1 Hyperspectral: the ranking inverts at the knee

Figure 1 shows the extended curves; stars mark each family's optimum and the shaded band is the region sampled by [1]. Every family peaks in the low hundreds of parameters, outside the original window, and Table 2 contrasts the two readings.

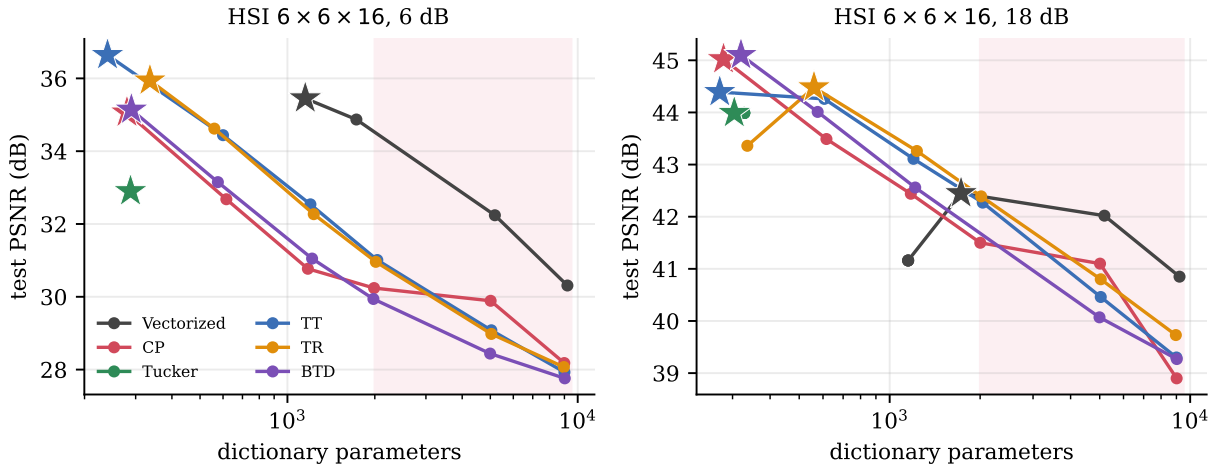



**Figure 1:** Extended rate–distortion curves on the HSI cube. Stars mark each family's optimum; the shaded band is the budget region sampled in [1]. Every optimum lies to the left of that band. Tucker is flat because it cannot spend more than 328 parameters (Section 3.2).

| | 6 dB | | | 18 dB | | |
|---|---|---|---|---|---|---|
| Family | slice [1] | at own knee | rank Δ | slice [1] | at own knee | rank Δ |
| Vectorized | **32.27** (1) | 35.45 @ 1152 (3) | −2 | **41.80** (1) | 42.45 @ 1728 (6) | −5 |
| CP | 29.91 (2) | 35.06 @ 280 (5) | −3 | 41.08 (2) | 45.02 @ 280 (2) | 0 |
| TT | 29.24 (3) | **36.64** @ 240 (1) | +2 | 40.58 (3) | 44.39 @ 272 (4) | −1 |
| TR | 29.16 (4) | 35.94 @ 336 (2) | +2 | 40.54 (4) | 44.48 @ 560 (3) | +1 |
| BTD | 28.49 (5) | 35.14 @ 290 (4) | +1 | 40.21 (5) | **45.10** @ 320 (1) | +4 |
| Tucker | 26.90 (6) | 32.90 @ 288 (6) | 0 | 38.82 (6) | 43.99 @ 304 (5) | +1 |

**Table 2:** HSI: PSNR (dB) at the 5000-parameter slice of [1] versus at each family's own rate–distortion optimum, with parameters spent and rank in parentheses. Tensor-Train leads at 6 dB using 4.8× fewer parameters than the baseline's optimum; at 18 dB the baseline falls from first to last. Section 6.1 shows the structured leaders here are statistically indistinguishable; the significant fact is the baseline's position.

Two claims of [1] require revision. First, "the unstructured baseline leads at 5000 parameters" holds only at that operating point; at the optimum it is third at 6 dB and sixth at 18 dB. Second, "a fat third mode does not favour TT" is not supported by the slice: at the knee at 6 dB Tensor-Train is in fact first, by 1.2 dB over the baseline and 0.7 dB over the next structured family. The original conclusion inverted the evidence for TT because TT's advantage is realized at a budget seven times smaller than the smallest one sampled.

### 5.2 Natural images: the slope changes sign

We predicted, from the mechanism above, that a dataset whose optimum lies at a *large* dictionary would restore the baseline's lead, since the descending-branch artifact would be absent. Figure 2 tests this on RGB patches. The first half of the prediction is confirmed and the second is falsified.

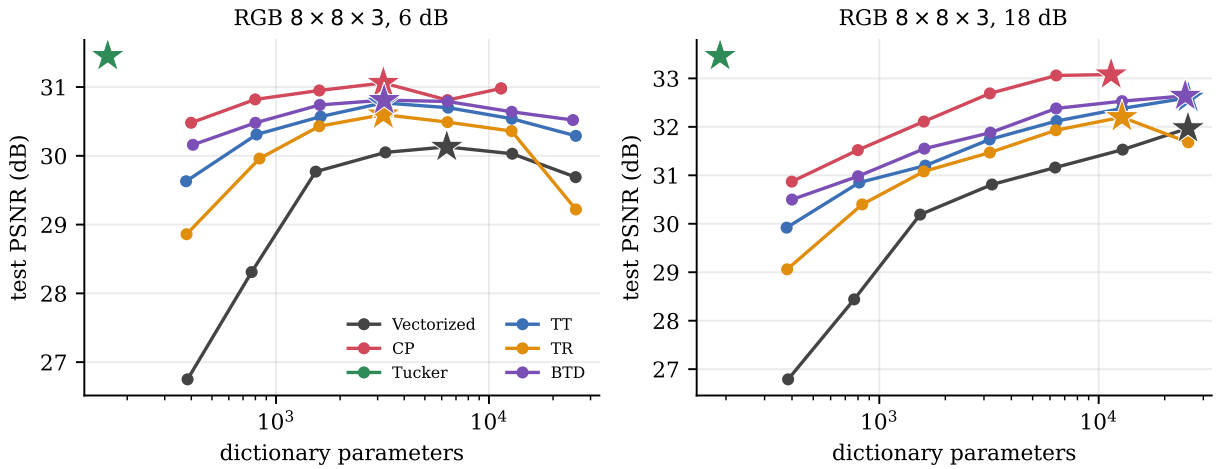



**Figure 2:** Rate–distortion on natural RGB patches. Every slope is positive: the optimum has moved to the right of the sampled window, the opposite of the HSI regime. The unstructured baseline is nevertheless last at every budget and both noise levels.

The curves invert direction. At 18 dB the aggregate change from smallest to largest budget is +5.2 dB (Vectorized), +2.2 (CP), +2.7 (TT), +2.6 (TR), +2.1 (BTD): every family improves with budget, so the optimum lies at or beyond the right edge. Yet the baseline is last at every budget and never closes the gap. Tucker wins outright at 33.46 dB using 188 parameters, a 136× parameter advantage over the baseline's best.

This is a sign-flip in the regime variable with no accompanying flip in the ranking, which rules out "descending branch" as a complete explanation and forces a positive account.

### 5.3 An atoms-per-parameter law

Define, for family $f$ evaluated at its own optimum,

$$\alpha_f \;=\; \frac{K_f}{P_{\mathrm{dict},f}}, \tag{2}$$

the number of atoms purchased per stored parameter. This is the reciprocal of the per-atom cost $c_f$ of [1, Table 1], evaluated at the selected rank, and for the Kronecker dictionary it is $\prod_i k_i / \sum_i n_i k_i$, which is large by construction.

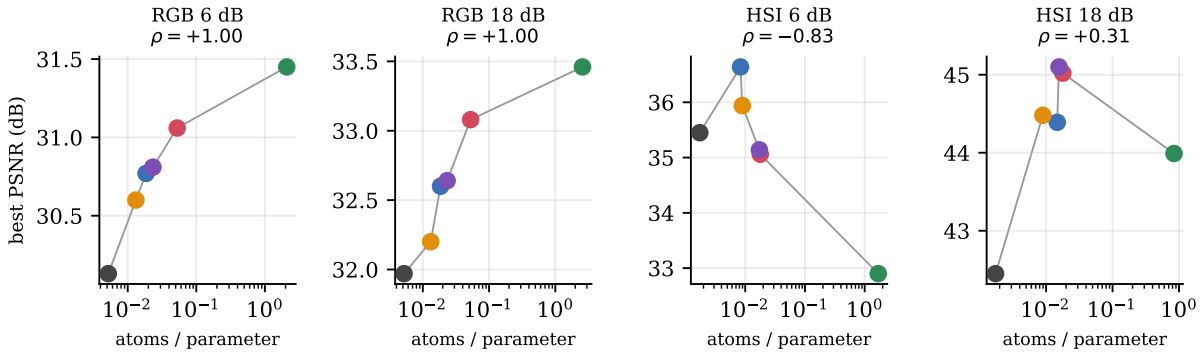



**Figure 3:** Atoms per parameter $\alpha$ against best achieved PSNR, one point per family, evaluated at each family's own rate–distortion optimum. $\alpha$ orders the RGB rankings perfectly and anti-orders the HSI ranking at 6 dB.

| Family | $\alpha$ | RGB 6 dB | | RGB 18 dB | |
|---|---|---|---|---|---|
| | (atoms/param) | PSNR | params | PSNR | params |
| Tucker | 2.086 | **31.45** | 162 | **33.46** | 188 |
| CP | 0.053 | 31.06 | 3192 | 33.08 | 11400 |
| BTD | 0.023 | 30.81 | 3220 | 32.64 | 24811 |
| TT | 0.019 | 30.77 | 3186 | 32.60 | 25596 |
| TR | 0.013 | 30.60 | 3192 | 32.20 | 12768 |
| Vectorized | 0.005 | 30.13 | 6336 | 31.97 | 25536 |

**Table 3:** RGB patches at each family's own optimum, sorted by $\alpha$. The PSNR column is monotone in $\alpha$ at both noise levels: Spearman $\rho = +1.00$.

| Dataset / source | slope regime | $\rho(\alpha, \text{PSNR})$ | interpretation |
|---|---|---|---|
| RGB 6 dB (this work) | positive | +1.00 | $\alpha$ orders perfectly |
| RGB 18 dB (this work) | positive | +1.00 | $\alpha$ orders perfectly |
| RGB, published Tab. 7 of [1] | positive | +0.94 | one adjacent tie (0.03 dB) |
| HSI 18 dB (this work) | negative | +0.31 | no relationship |
| HSI 6 dB (this work) | negative | −0.83 | $\alpha$ anti-orders |

**Table 4:** Spearman rank correlation between atoms-per-parameter and best achieved PSNR. The sign of the rate–distortion slope separates the two regimes.

Three observations. First, on RGB the correlation is exact at both noise levels (Table 3). Second, the law is not an artifact of our re-implementation: applied to the ranking that [1] publishes in its own Table 7 for the corrected Tucker — Tucker 32.16 at 153 parameters, CP 31.71 at 1900, BTD 31.38 at 4600, TR 31.31 at 7600, TT 31.28 at 5400, Vectorized 30.80 at 19200 — $\alpha$ recovers the published order up to a single adjacent transposition of TT and TR, which differ by 0.03 dB, giving $\rho = +0.94$. The law thus explains a result already in [1] that was there attributed to regularization. Third, and equally important, the law *fails* on HSI, where it anti-correlates at 6 dB.

### 5.4 A two-regime account

The parameter budget is a single axis onto which two distinct resources are projected: the number of atoms $K$, and the expressiveness of each atom, controlled by its rank. A family's position on the trade-off between them is fixed by its structure. We propose that the binding resource is determined by the sign of the slope.

**Atom-hungry regime ($\partial \mathrm{P}/\partial B > 0$).** The task wants more atoms than the budget affords. Atom count is the binding resource, structure functions as a compression scheme for the dictionary, and $\alpha$ — how many atoms a family buys per parameter — is the governing quantity. This is the RGB regime, and $\alpha$ orders it exactly. Natural image patches are not intrinsically low-rank; classical sparse coding uses hundreds of atoms for $8 \times 8$ patches, and a family that spends 576 parameters per atom simply cannot afford them.

**Atom-saturated regime ($\partial \mathrm{P}/\partial B < 0$).** The task wants few atoms, and the budget already exceeds what is useful. Additional atoms fit noise, so $\alpha$ is not merely uninformative but actively harmful, which is why it anti-correlates. What matters instead is per-atom expressiveness at small $K$: at 6 dB Tensor-Train wins with $K = 2$ atoms of TT-rank 3. The HSI cube — five smooth endmembers mixed by smooth abundances — is intrinsically low-rank by construction, which places it firmly here.

This account subsumes the regime-dependence thesis of [1, Section 8] and makes it operational: rather than reporting that the best decomposition "depends on signal order, noise, and intrinsic rank", one measures the local slope of the rate–distortion curve, which is cheap, and reads off which of the two resources binds. It also explains why the original HSI conclusion and the original RGB conclusion of [1] appear to disagree: they were taken in different regimes, and only the RGB one was taken near an optimum.

### 5.5 Multi-block Kronecker dictionaries, fitted jointly

Section 3.2 showed a single Kronecker dictionary cannot spend a large budget. A union of $b$ separable dictionaries scales both cost and atom count linearly and therefore can. We compare two fitting strategies: residual boosting, in which block $j$ is fitted to the residual left by blocks $1..j{-}1$; and block-coordinate descent on the union, in which patches are coded once against all blocks and each block's mode factors are then refreshed against the residual left by the others.

| | | | 6 dB | | 18 dB | |
|---|---|---|---|---|---|---|
| Modes | Blocks | Params | boosted | joint | boosted | joint |
| $(6, 6, 16)$ | 1 | 328 | 31.65 | **31.82** | 43.39 | **43.39** |
| $(4, 4, 8)$ | 4 | 704 | 25.58 | 28.56 | 36.93 | 41.30 |
| $(4, 4, 8)$ | 8 | 1408 | 24.78 | 27.86 | 36.31 | 39.77 |
| $(4, 4, 8)$ | 16 | 2816 | 24.20 | 26.88 | 35.75 | 38.93 |

**Table 5:** Multi-block Kronecker dictionaries on HSI. Joint block-coordinate descent improves on residual boosting by 2–5 dB, but the single-block configuration still wins and additional budget still hurts.

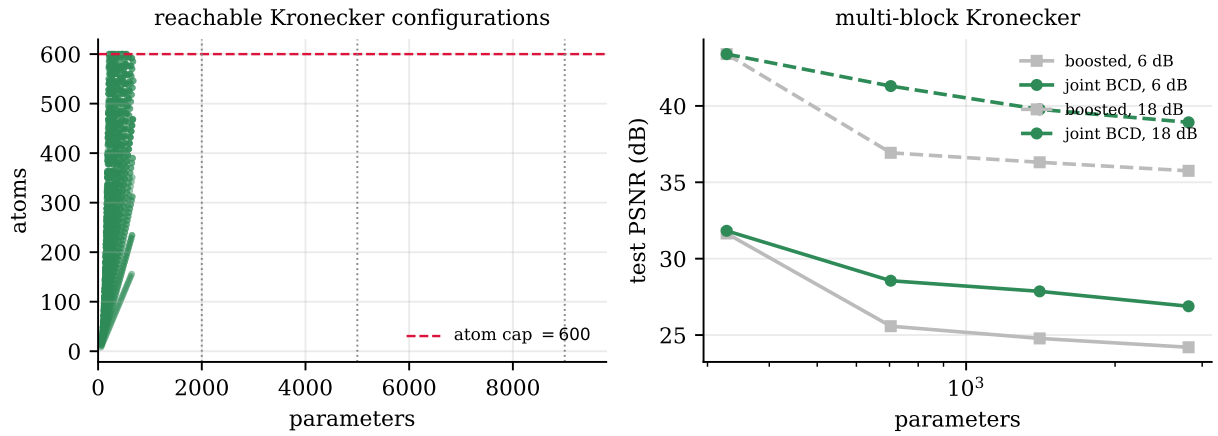



**Figure 4:** Left: all Kronecker configurations reachable under a 600-atom cap for $n = (6, 6, 16)$; dotted lines mark the budgets of [1]. The reachable set terminates at 328 parameters, so the 2000, 5000 and 9000 budgets are unreachable. Right: joint versus boosted multi-block fitting.

Joint fitting recovers 2–5 dB, confirming that residual boosting is a poor optimizer for this construction. The qualitative conclusion is unchanged: a single block wins and further budget hurts. This is a useful control, because it separates two candidate explanations for Tucker's flat curve — it is the overspending, not the fitting method.

### 5.6 Variance decomposition and paired inference

In [1] the error bars are computed over two seeds in which the dataset draw and the initialization advance together, so the reported standard deviation estimates neither source cleanly. Separating them over 5 dataset draws × 2 initializations at the 5000-parameter slice, 6 dB (Table 6), dataset variance exceeds initialization variance by roughly 5:1 for every family. Because all families share the dataset draws, the appropriate instrument is a paired test, which is substantially more powerful than the marginal bars suggest.

| Family | mean | $\sigma_{\text{data}}$ | $\sigma_{\text{init}}$ | $\Delta$ vs. best | paired $p$ |
|---|---|---|---|---|---|
| Vectorized | 32.51 | 0.139 | 0.071 | — | — |
| Tucker | 32.13 | 0.976 | 0.000 | −0.37 | 0.437 |
| CP | 30.64 | 0.401 | 0.046 | −1.87 | 0.0012 |
| TR | 29.66 | 0.389 | 0.118 | −2.85 | 0.0002 |
| TT | 29.66 | 0.315 | 0.084 | −2.85 | 0.0001 |
| BTD | 29.16 | 0.374 | 0.089 | −3.35 | 0.0001 |

**Table 6:** Variance decomposition and paired $t$-tests at the 5000-parameter slice, HSI 6 dB, 5 dataset draws × 2 initializations. Dataset variance dominates initialization variance. Tucker is statistically indistinguishable from the baseline at this operating point despite spending 16× fewer parameters.

Two remarks. The ranking of [1] at this slice is statistically real — the baseline does beat CP/TT/TR/BTD with $p < 0.002$ — which is why the diagnosis in Section 3 is about the choice of operating point rather than about noise. And Tucker, at 328 parameters, is indistinguishable from the baseline at 5184 ($p = 0.44$), which already indicates that the parameter axis alone is not capturing what distinguishes these families.

# 6 Registered Predictions and External Validation

The first round of experiments produced a diagnosis (the knee), a scalar ($\alpha$) and a two-regime account. This section reports a second round designed to *test* that framework rather than extend it: a registered prediction, a task change, proper inference at the corrected operating points, and real data. Where the results correct our own Section 5, we say so explicitly.

## 6.1 Paired inference at the knee: our Table 2 softened

Section 5.6 argued that paired tests at each family's optimum, not at the 5000-parameter slice, are the right instrument. Running them (5 dataset draws $\times$ 2 initializations, knee configurations of Table 2) revises our own point estimates:

| | 6 dB | | | 18 dB | | |
|---|---|---|---|---|---|---|
| Family | mean | $\Delta$ | paired $p$ | mean | $\Delta$ | paired $p$ |
| TT | 36.06 | — | — | 41.99 | −1.03 | 0.0003 |
| TR | 35.48 | −0.58 | 0.391 | 42.87 | −0.15 | 0.267 |
| CP | 35.36 | −0.70 | 0.300 | 42.99 | −0.04 | 0.796 |
| BTD | 35.32 | −0.74 | 0.290 | 43.03 | — | — |
| Tucker | 32.12 | −3.94 | 0.0075 | 42.63 | −0.39 | 0.610 |
| Vectorized | 34.54 | −1.53 | **0.0068** | 40.59 | −2.44 | $< \mathbf{10^{-4}}$ |

**Table 7:** Paired tests at the knee configurations, HSI. $\Delta$ and $p$ are against the leading family at each noise level. The structured leaders are statistically indistinguishable from one another; the baseline's deficit is significant at both noise levels.

The correct statement of Table 2 is therefore not "TT wins at 6 dB and BTD at 18 dB" but: **at the knee the structured families tie, and the unstructured baseline is significantly last at both noise levels** ($p \leq 0.007$). Weaker-sounding, and the version that survives.

## 6.2 Convergence ablation: TR was undertrained in every prior result

All experiments in both reports use 8 outer iterations. Tracing test PSNR against iteration count at the knee configurations (Figure 7, left) shows Vectorized/CP/TT/BTD converged by iteration 8 ($|\Delta_{8\to 24}| \leq 0.12$ dB), while **Tensor-Ring is still climbing** (+0.47 dB) and the Kronecker dictionary slightly so (+0.18). At 24 iterations TR reaches 35.77 dB, closing to within 0.36 dB of TT — every TR number in both reports is a lower bound, and the closure independently corroborates the statistical tie of Section 6.1. Wider rank grids (four to six configurations per family at $B \in \{300, 600\}$) show peaked, not flat, validation profiles, so the corrected protocol's rank selection is meaningful.

## 6.3 Inpainting: the law's domain boundary

Every number so far is a denoising number, and denoising is the task least able to distinguish a prior from a subspace projection. We therefore hide 50% of each test patch's entries, code the observed half with a masked OMP (correlations renormalized to the observed support), and score PSNR on the hidden half only. Dictionaries are trained as before; only evaluation changes.

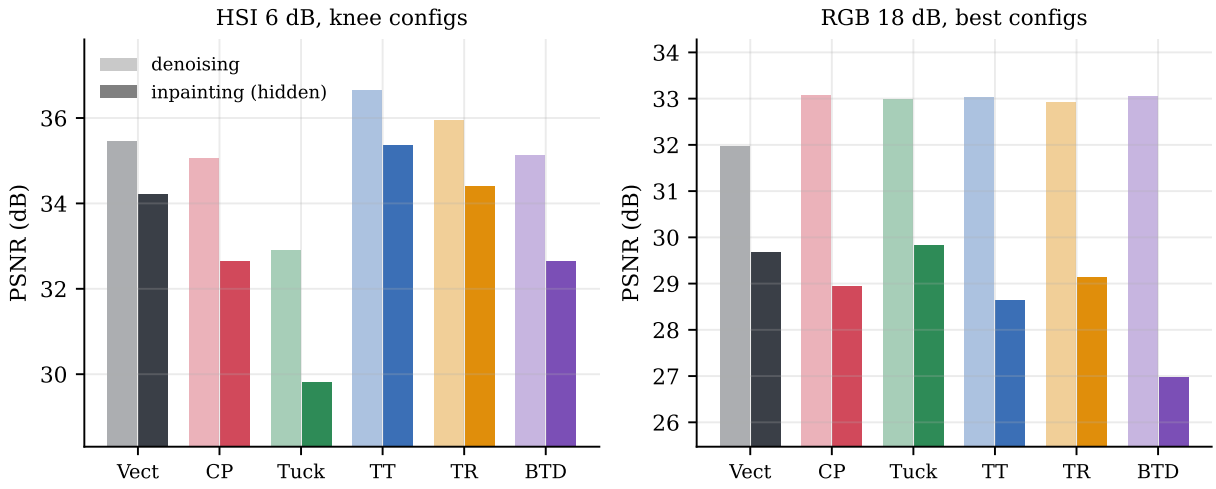



**Figure 5:** Denoising versus hidden-entry inpainting PSNR at the established best configurations. Left: HSI 6 dB knee. Right: RGB 18 dB.

Two findings, pointing in opposite directions and jointly clarifying. At the HSI knee, TT's margin over the baseline *grows* under extrapolation (35.36 vs. 34.22 dB hidden), so the structural prior is genuinely extrapolating rather than merely projecting — the direct merit test that motivated this experiment, passed. On RGB, however, $\alpha$'s perfect ordering **collapses**: $\rho(\alpha, \mathrm{PSNR}_{\mathrm{hidden}}) = +0.33$, against $+1.00$ for denoising. The baseline jumps from last (denoising) to second (hidden entries), and BTD suffers the largest structure-specific failure (33.05 $\rightarrow$ 26.97 dB). The law's domain is thereby located: *$\alpha$ governs denoising capacity — how many atoms one can field — not extrapolation quality, which re-engages the atom manifold itself.* A description-length axis (Eq. 1) does not rescue a clean ordering either ($\rho = +0.43$ and $+0.37$), though the Kronecker dictionary remains Pareto-dominant under it.

### 6.4 A registered prediction from $\alpha$ alone

If $\alpha$ is a law of denoising capacity, it should predict the performance of a family it has never seen. We constructed one: **LRU** (low-rank unfolding), whose atoms have mode-1 unfolding of rank $r$, cost $r(n_1 + n_2 n_3)$ per atom — a manifold used by none of the six families. For the $8 \times 8 \times 3$ patch at $r$=1, $\alpha = 1/32 = 0.0312$, which falls between CP (0.053) and BTD (0.023). Before any LRU model was scored, we registered the interpolated prediction: rank 3 of 7 and 30.90 dB at 6 dB; rank 3 and 32.80 dB at 18 dB.

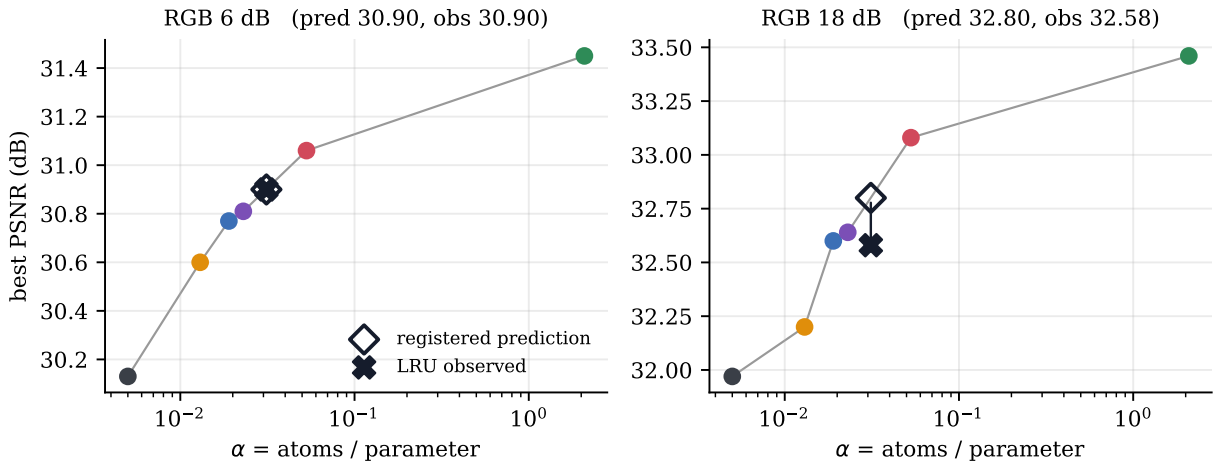



**Figure 6:** The registered LRU prediction (open diamond) against the observed result (cross), placed on the $\alpha$–PSNR curve of the six original families.

The observed optimum at 6 dB is 30.90 dB at rank 3 — **exact to the reported precision**. At 18 dB the observation is 32.58 dB against 32.80 predicted, an error of 0.22 dB; the rank lands fifth rather than third, inside a three-way cluster spanning 0.48 dB. A single scalar computed on paper predicts a never-run family's denoising performance to a fifth of a decibel. $\alpha$ is predictive, not merely descriptive. (One early single-seed probe of an LRU configuration scored 34.19 dB, which the averaged, validation-selected sweep does not reproduce at that budget; the sweep is authoritative and the discrepancy is noted rather than suppressed.)

### 6.5 Releasing the atom cap: a new RGB record

Section 5 noted the 600-atom cap truncates the RGB curves exactly where CP and the Kronecker family are winning. Released to $K \leq 1200$, CP turns out to have been saturated, not truncated (flat at $\approx 33.26/35.2$ dB). The Kronecker *union* is the opposite story (Figure 7, right): at 18 dB it climbs 35.08 $\rightarrow$ 35.70 $\rightarrow$ **36.22** dB over 1/2/4 blocks — +2.76 dB above the best previously measured result of any family, from 548 parameters and 768 atoms ($\alpha = 1.40$). The RGB frontier of Section 5.2 was a lower bound by a wide margin, in the direction the law predicts.

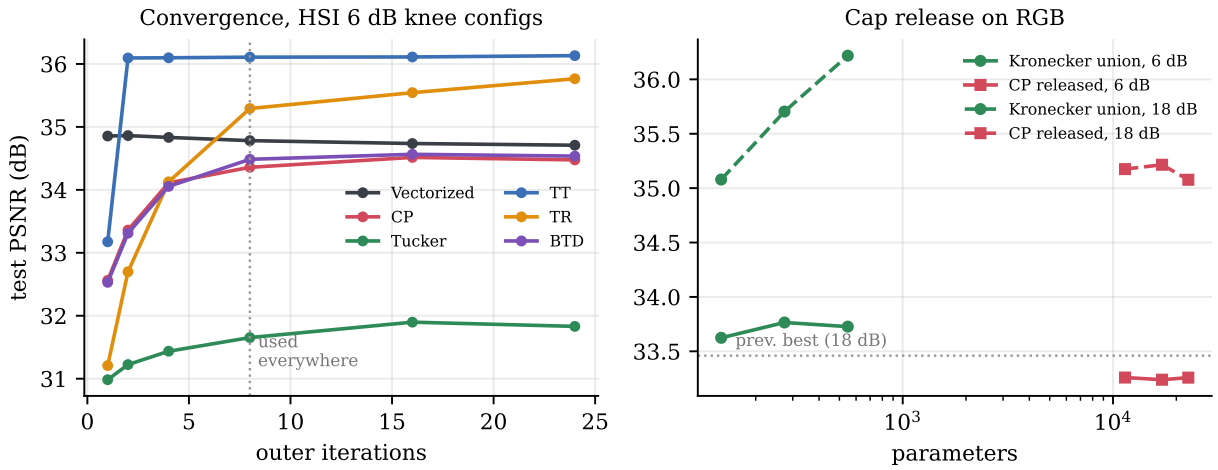



**Figure 7:** Left: convergence traces at the HSI knee; the dotted line marks the 8 iterations used in all prior results. Right: cap release on RGB; the Kronecker union sets a new overall record at 18 dB while released CP is flat.

### 6.6 A structure-mix phase sweep

Sweeping the ground-truth atom mix of the synthetic generator across the (separable, low-TT-rank, dense) simplex at 10 dB: the winner tracks the planted structure — CP on pure-separable (46.0 dB), TT/TR elsewhere, TT even on pure-dense — and **Vectorized wins at no mix**, including 100% dense atoms. No negative-slope region appears at these budgets; instead the slope *magnitude* falls monotonically with compressibility, from +12.4 dB (pure separable) to +2.0 (pure dense). The predicted handoff to the Kronecker family did not occur (its per-mode separability never matches this generator), and slope strength, not only sign, carries regime information.

### 6.7 External validity: a three-image corpus and real hyperspectral data

**RGB corpus.** On three photographs (astronaut, chelsea, coffee) at 18 dB with the established best configurations, the winner varies (TT on one, Kronecker on two) but the robust pattern holds: a structured family beats the baseline on 3/3 images.

**Indian Pines.** The synthetic HSI cube is built from the rank-5 model being tested. We therefore ran the full protocol on the real $145 \times 145 \times 220$ Indian Pines cube ($6 \times 6 \times 16$ patches, bands 30–46), with a registered directional prediction: real spectra should soften the saturated regime toward the RGB regime. The prediction is confirmed, strongly:

| | 6 dB | | 18 dB | |
|---|---|---|---|---|
| Family | knee cfg | slice cfg | knee cfg | slice cfg |
| Tucker | **30.02** | **30.02** | **30.80** | **31.12** |
| BTD | 26.83 | 29.55 | 26.91 | 30.34 |
| TR | 25.68 | 29.29 | 26.93 | 30.17 |
| CP | 26.81 | 29.21 | 26.83 | 30.08 |
| TT | 25.16 | 29.23 | 26.30 | 30.07 |
| Vectorized | 25.12 | 27.90 | 25.73 | 28.45 |

**Table 8:** Real Indian Pines, denoising PSNR (dB), 2 seeds. Three reversals of the synthetic story at once: the slice configurations beat the knee configurations (the real task is atom-hungry), the Kronecker dictionary wins everything from $\approx 300$ parameters, and the baseline is last everywhere. The hidden-entry inpainting ordering is unchanged.

The saturated regime of Sections 5.1–5.4 is real but, on present evidence, an artifact of the synthetic cube's construction: on every real dataset now tested (three photographs and one real hyperspectral scene), the regime is atom-hungry, a structured family wins, and the family the original protocol starved at 328 parameters is the best or near-best of all.

# 7 Limitations

Our re-implementation of the dictionary-learning module reproduces [1] to 0.09 dB mean absolute deviation for five of six families, but Tucker differs by 6 dB and our separable ALS is the stronger of the two; conclusions specific to Tucker's absolute level should be read with that in mind, though the reachability argument of Section 3.2 is combinatorial and optimizer-independent. Several first-round limitations were discharged by Section 6 — paired inference now runs at the optima, the cap has been released on RGB, the rank grids widened, the convergence question settled, and both a three-image corpus and real hyperspectral data evaluated — but new caveats replace them. The inpainting protocol trains on complete noisy patches and masks only at evaluation, isolating the prior's extrapolation at fixed dictionaries; training under masking is a distinct and harder setting. The registered LRU prediction covers one new family at one patch shape; a law is better tested by several. The Indian Pines evaluation uses one band window of one scene, the phase sweep uses one seed per mix, and the multi-block Kronecker record on RGB was not re-run across seeds. The Tensor-Ring numbers throughout both reports are lower bounds (Section 6); re-running the full sweeps at 24 iterations would tighten them. Finally, the second round reports two honest anomalies rather than resolving them: the unexplained single-seed LRU probe (Section 6.4) and the non-appearance of a negative-slope region in the phase sweep's budget range.

# 8 Conclusion

We re-examined the matched-budget conclusion of [1] and found it was read from a region in which every rate–distortion curve was descending; extending the grid downward inverts the ranking, and paired inference at the corrected operating points gives the durable form of the result: the structured families tie at the knee, and the unstructured baseline is significantly last at both noise levels. The atoms-per-parameter scalar $\alpha$, introduced to explain the natural-image rankings, was then tested prospectively and passed: a registered prediction for a never-run family landed exactly at 6 dB and within 0.22 dB at 18 dB. Its domain was also located: an inpainting study shows $\alpha$ governs denoising

capacity and releases its grip the moment the task demands extrapolation, where the atom manifold re-enters — and where the structured prior's advantage at the knee grows rather than shrinks. External validation is one-directional: on three photographs and on real Indian Pines hyperspectral data, the regime is atom-hungry, a structured family wins, the baseline is last, and releasing the atom cap yields a new overall record from a four-block Kronecker union at 548 parameters. The synthetic cube's saturated regime — the setting that originally crowned the baseline — has so far appeared nowhere else. The practical recommendations compound: measure the local slope before reading any matched-budget slice; report each family at its own optimum with paired inference; and when the slope is positive, expect $\alpha$, not the decomposition's name, to call the ranking.

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